MIL-C-36033A
15 August 1967
SUPERSEDING
MIL-C-36033
25 May 1960

MILITARY SPECIFICATION

CHESTS, MEDICAL INSTRUMENT AND SUPPLY SETS, RIGID, EMPTY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight, watertight, aluminum chests used to contain and protect various items of medical material in the field.

1.2 Classification. Rigid medical field kit chests shall be of the following sizes, as specified (see 6.2):

Size No.
1
2
3

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 6545

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SPECIFICATIONS

Federal

QQ-A-225/8	Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished, 6061.
QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet.
QQ-W-423	Wire, Steel, Corrosion-Resisting.
TT-E-527	Enamel, Alkyd, Lustreless.
MMM-A-250	Adhesive, Water Resistant.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure Sensitive Adhesive Paper, Water Resistant.

Military

MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-S-6855	Synthetic Rubber Sheets, Strips, Molded or Extruded Shapes.
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals; Test Methods.
Fed. Std. No. 595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MS 16686	Medical Services Symbol.

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DRAWING

Defense Personnel Support Center

24084 Chest, Medical Instrument and Supply Set, Rigid, No. 1, 2, and 3, Empty.

(Copies of specifications, standards, drawing, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First Article. When required (see 6.2), prior to the start of production of the contract quantity, samples of the chests shall be assembled, completely finished, and tested by the contractor. The samples shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first articles, acquisition of materials, components, or commencement of production is at the sole risk of the contractor. Accepted sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the sample is not approved, the contractor shall submit a new sample for approval, or modify the existing sample to remove the deficiencies cited.

3.2 Component parts. Each chest consists of a body, a cover, the necessary drawbolts for securing the cover to the body, and a gasket which is retained in the cover and against which the body bears to make the waterproof joint. Chests (size No. 2 and 3) each have a handle secured to the body by a handle plate. Chest (size No. 3) has a partition. Each drawbolt consists of a draw, a strike, a swivel plate, a draw ring, and a hinge pin.

3.3 Materials. Except as specified herein, materials shall conform to the applicable portions of the specifications, standards, and drawing listed in section 2, and shall be capable of meeting the requirements shown therein. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in 4.5.3. When materials to be used are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended.

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3.3.1 Aluminum alloys.

3.3.1.1 Allumimum alloy sheet for the chest body, cover, body partition (chest, size No. 3), and handle plate (chests, size No. 2 and 3) shall be alloy designation 5052, temper O, conforming to QQ-A-250/8.

3.3.1.2 Aluminum alloy sheet for the drawbolt swivel plate, drawbolt strike, and drawbolt draw shall be alloy designation 5052, temper H24 or H34, conforming to QQ-A-250/8.

3.3.1.3 Aluminum alloy rod for the handle (chests, size No. 2 and 3) shall be alloy designation 6061, temper T4 conforming to QQ-A-225/8.

3.3.2 Corrosion-resisting steel wire. The corrosion-resisting steel wire shall conform to Form I of QQ-W-423.

3.3.2.1 Wire for the drawbolt draw ring shall conform to composition 302, condition B of QQ-W-423.

3.3.2.2 Wire for the drawbolt hinge pin shall conform to composition 316, condition B of QQ-W-423.

3.3.3 Rubber. Rubber for the gasket shall conform to class 3, grade 30 of MIL-S-6855. The color of the rubber is optional. The rubber shall be tested as specified in 4.5.1. The physical properties shall be as follows:

Minimum tensile strength	1000 p.s.i.
Minimum elongation	450 percent
Maximum compression set	40 percent
Durometer reading	30 ± 5
Maximum loss in tensile strength on accelerated aging	20 percent
Maximum loss in elongation on accelerated aging	40 percent
Maximum change in durometer reading on accelerated aging	+15, -0
Maximum loss in weight on accelerated aging	10 percent

3.3.4 Primer. The primer (if used) shall conform to MIL-P-8585, and shall be color Y of the same specification.

3.3.5 Enamel. Enamel shall conform to TT-E-527. Color shall be green, color X-34087 of Fed. Std. No. 595, except that for marking as specified in 3.8, the color shall be black, color 37038 of the same standard (see 6.3).

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3.4 Construction. Construction shall conform to the requirements specified herein and to those shown in the drawing. The construction shall be sound, durable, and capable of meeting all tests hereinafter specified.

3.4.1 Gasket. The rubber gasket shall be molded of the material specified in 3.3.3 to form a seamless, endless unit which conforms in shape, contour, and dimensions of the drawing. The gasket shall be fitted into the retainer in the cover after baking, specified in 3.6, has been accomplished and the cover has cooled. The surfaces of the gasket, against which the body bears, shall be suitably treated to prevent adhesion. The gasket shall capable of rendering the chest watertight and shall be suitable for sealing the chest during normal use, which includes many openings and closings.

3.4.2 Handle and handle plate. Handle and handle plate (chests, size No. 2 and 3) shall be die-formed.

3.5 Welding. Welding shall be done by qualified welders using appropriate equipment in a qualified process. Maximum distance between spot weld centers shall be two inches where two or more spot welds are used to attach one piece to another. Spot welds shall be capable of developing the tensile strength of the parent metal. All exposed spot welds shall be smoothed sufficiently and shall not be visible after the enamel finish as specified in 3.6 has been applied.

3.6 Finish. After assembly has been completed, except for insertion of the gasket as specified in 3.4.1, all accessible aluminum alloy surfaces of the chest, both inside and outside, shall be finished with enamel as specified herein. Surfaces to be enameled shall be thoroughly cleaned and degreased, then prepared for topcoating by one of the following three methods:

(1) Chemical treatment in accordance with MIL-C-5541.

(2) Anodic treatment in accordance with MIL-A-8625.

(3) Priming treatment, using material specified in 3.3.4, thinned and applied by spray in accordance with the directions specified under "Preparation for Delivery" in MIL-P-8585, and air-dried for not less than 30 minutes before topcoating.

Topcoating shall be with one coat of enamel as specified in 3.3.5, reduced as specified in the paragraph titled "Spraying properties" in the requirements section of TT-E-527, applied by spraying and cured by a heat treatment equivalent to baking at 250° Fahrenheit (F.) for not less than 45 minutes. The resultant finish shall be hard, smooth, and free from sags and runs. Contractor shall endeavor to control film thickness so that the thickness of a single dried coat is 0.25 to 0.35 mil for primer (if used) and 1.0 to 1.5 mil for enamel.

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3.7 Watertightness. The finished chest, including the gasket, shall be tested as specified in 4.5.2. There shall be no entry of water into the chest.

3.8 Marking for identification. The exterior base of the chest body shall be centrally marked with a medical services symbol in accordance with type I or type III of MS 16686. The letters "US" shall be a minimum of 1/2 inch high. The name of the manufacturer or registered trademark, and the contract number or purchase order number shall be marked or stamped on the inside of the lid of each chest or on the exterior base of the chest body. When marked on the base, the center of the identification marking shall be 2-1/2 inches to the right of, and 1-1/4 inches below the center of the medical services symbol. Letters and numerals shall be of uniform height, 5/32 to 5/16 inches. Marking for identification shall be accomplished with black lustreless enamel as specified in 3.3.5, and applied by the silk screen or equally suitable method so that there shall be no bleeding, or shall be metal stamped. Marking shall remain clearly legible during normal use and resist rubbing off. Metal stamping shall be distinct and legible.

3.9 Workmanship. Workmanship shall be first class throughout. Chests shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 First article inspection. When required, examination and testing of the first article sample shall be made on one or more completely fabricated chests for all provisions of this specification applicable to end product examination and tests.

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

	Inspection Level	AQL (percent defective)	Minimum sample size
Component material testing.	S-1	*	2
End item testing (water-tightness).	S-1	*	3

* Acceptance number shall be zero for all sample sizes.

4.3.3 For chemical analysis of metals. Four ounces of each lot of each metal used in the construction of the chests shall be selected for chemical analysis as specified in 4.5.3.

4.3.4 Component materials. Sufficient samples of the component materials shall be selected from each lot of material used in the manufacture of the chests to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for component materials covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated in 4.3.2.

4.3.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.4 Examination. The chests shall be examined to determine compliance with all requirements contained in this specification.

4.5 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.5.1 Rubber. The rubber for the gaskets shall be tested in accordance with the applicable requirements of MIL-S-6855. The "As-received" tests shall include hardness, tensile strength and elongation, and specific gravity. The tests following the specified accelerated aging shall include change in hardness, change in tensile strength and elongation, change in weight, and compression set.

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4.5.2 Watertightness. There shall be placed in an otherwise empty chest, sufficient weight to insure that the closed chest will sink in water. The chest shall be closed at room temperature, and the closed chest placed in a suitable container with the cover horizontal and uppermost. Water at the temperature of melting ice shall be placed in the container to a level two feet above the chest. The water level and temperature shall be maintained for four hours. Thereafter, the chest shall be removed from the water, the outside carefully blotted dry, and the chest then opened to determine whether any water had entered.

4.5.3 Chemical analysis of the metals. The samples selected in accordance with 4.3.3 shall be analyzed in conformance with method 111 or 112 of Federal Test Method Standard No. 151. In case of dispute, method 111 shall be used for referee analysis.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One chest, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging quantities

Size No.	Unit package	Intermediate package	Exterior container
1	1 unit	Not required	24 units
2	1 unit	Not required	12 units
3	1 unit	Not required	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with two inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The chests covered by this specification are intended to be used as containers for various assemblies of medical material to provide medical field kits of three sizes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Chest size number required (see 1.2).

c. When first article is required (see 3.1).

d. Where first article shall be sent (see 3.1).

e. Packaging and packing levels (see 5.1 and 5.2).

6.3 Availability of color chips. The individual three by five inch chips of the colors referenced in 3.3.5 are available from Commanding Officer, U. S. Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 This specification does not include all types, classes, grades, or sizes, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Code: N

Project Number: 6545-1006

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CAUTION

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DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-36033A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 54, WHICH MAY BE USED.

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